



# DataChallenge Raidium

Medical Segmentation of Anatomical Structures with  
Missing Labels

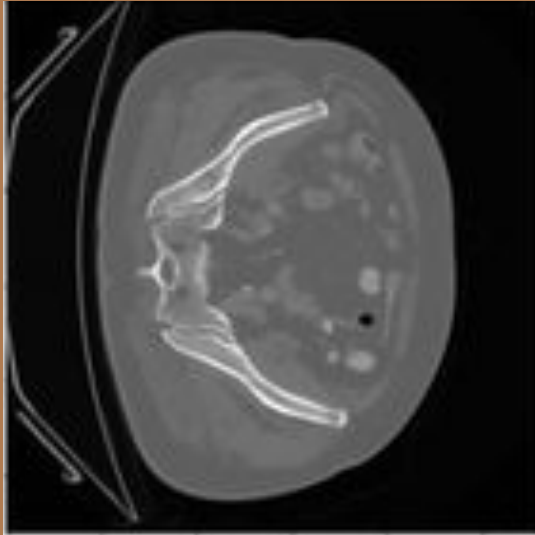
Presented by : Darius Dabert & Adrien Tousnakhoff



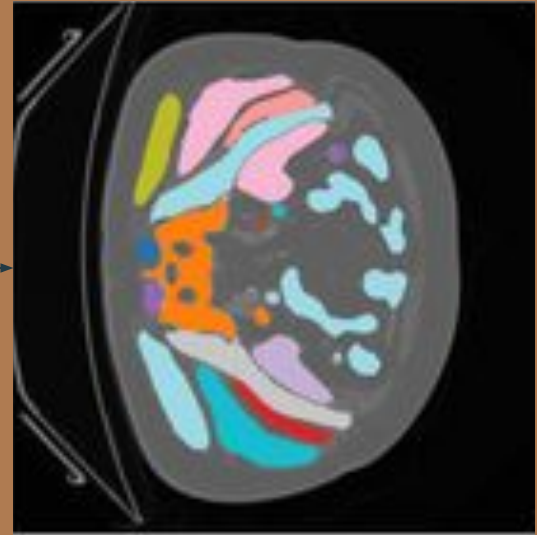
# Introduction – Context & Motivation

- **Context:** High-resolution CT scans provide detailed 3D images of the human body, enabling precise anatomical analysis.
- **Motivation:** Inconsistent and incomplete annotations across datasets make semantic segmentation challenging.
- **Goal:** Develop a model that segments anatomical structures based on shape, despite missing annotations.

# Problem Definition



Segmentation



Segmentation task on CT-Scans

# Problem Definition

- **Core Challenge:**

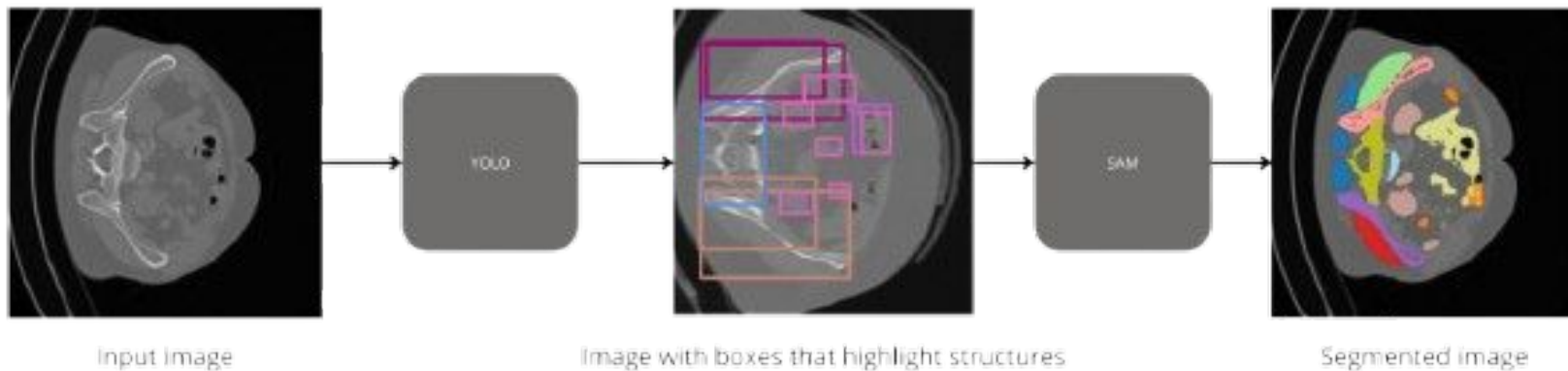
A lot of the data available for us is not annotated or only partially segmented.

- **Proposed Solution:**

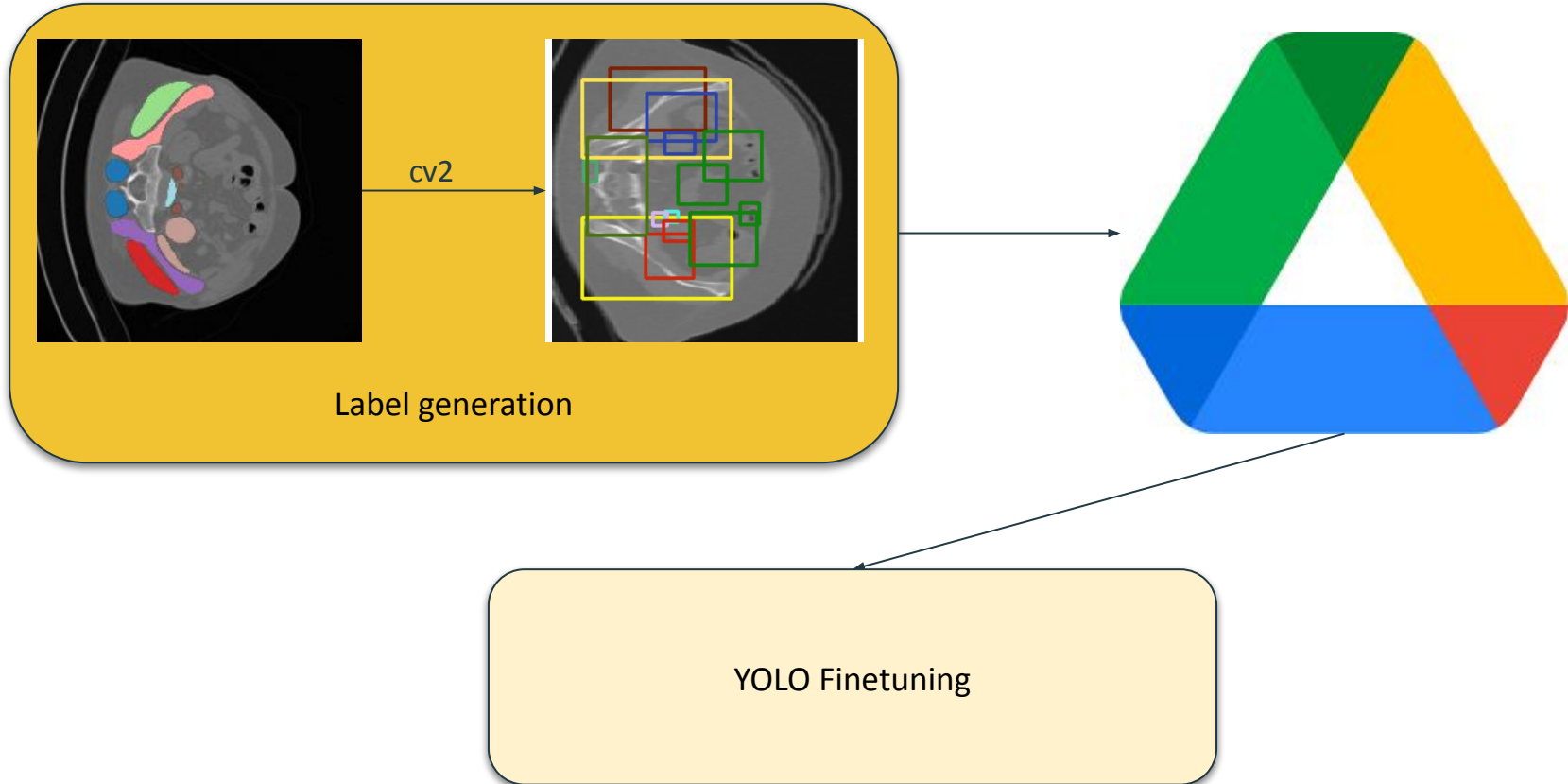
Use models pre trained on vast amount of data that only need small fine tuning to perform well.

→ SAM and YOLO

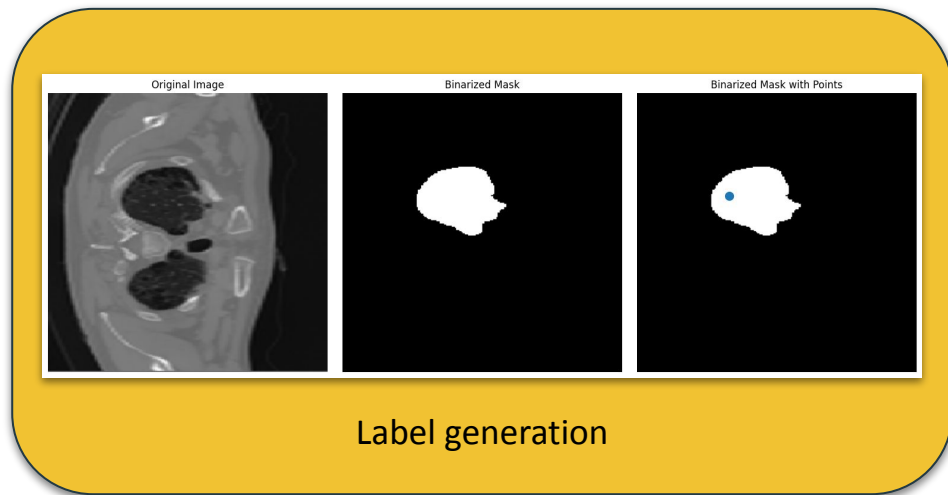
# Method Overview – Segmentation Pipeline



# Method Overview – YOLO finetuning



# Method Overview – SAM Finetuning



SAM Finetuning with BCELoss and IoU

# Method Overview – Using non-annotated data

- **Free Match Algorithm**

Train model both on labeled and unlabeled training set.

- **Self-supervised learning ( Not implemented )**

Inspired by the work at Meta on Self-Supervised Learning : DINO.



# Method Overview – Free Match

We create batches that contain labeled data as well as unlabeled data.

- **Labeled samples:**

For the labeled samples we proceed as usual, we compute predictions and backpropagate the loss.

- **Unlabeled samples:**

For the unlabeled samples, we compute a prediction and an associated confidence value. If the confidence exceeds a threshold, the prediction is considered as a label and we train on an augmented version of the image.

# Method Overview – Free Match

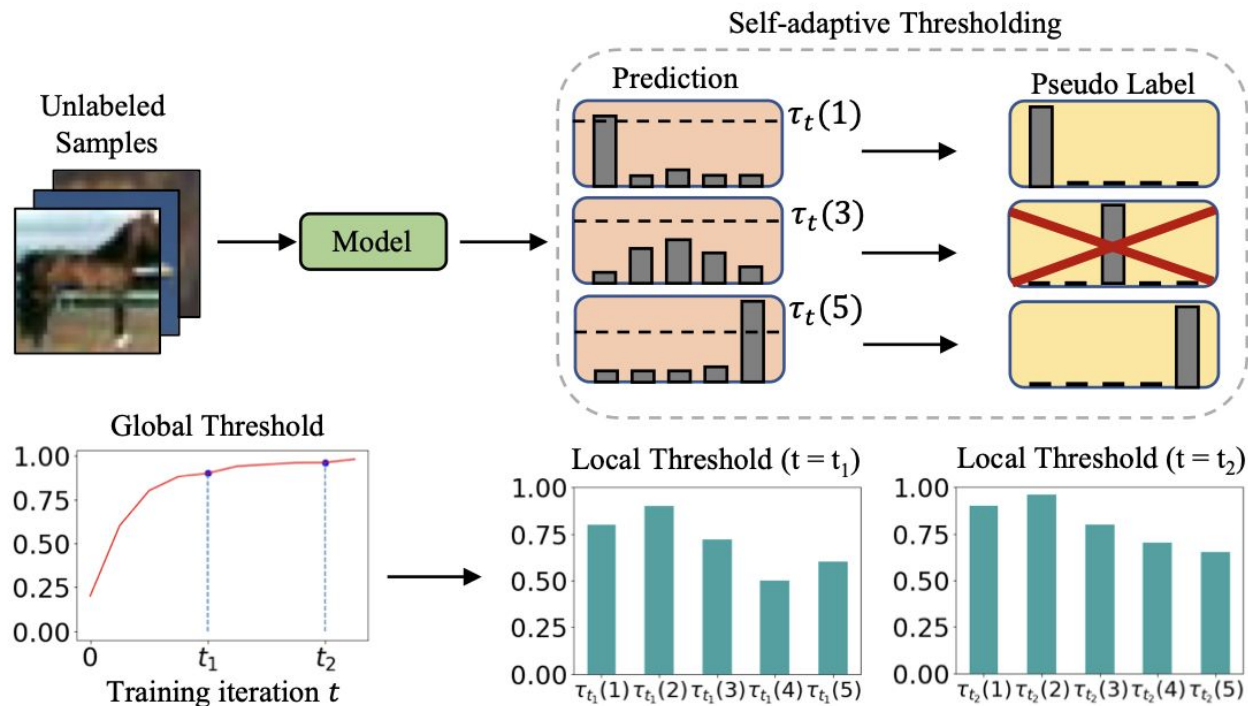


Figure 2: Illustration of Self-Adaptive Thresholding (SAT). FreeMatch adopts both global and local self-adaptive thresholds computed from the EMA of prediction statistics from unlabeled samples. Filtered (masked) samples are marked with red X. (figure from FreeMatch Paper)

# Method Overview – Self Supervised Learning

- **Data augmentation:**

For each image we compute different augmentations of the image.

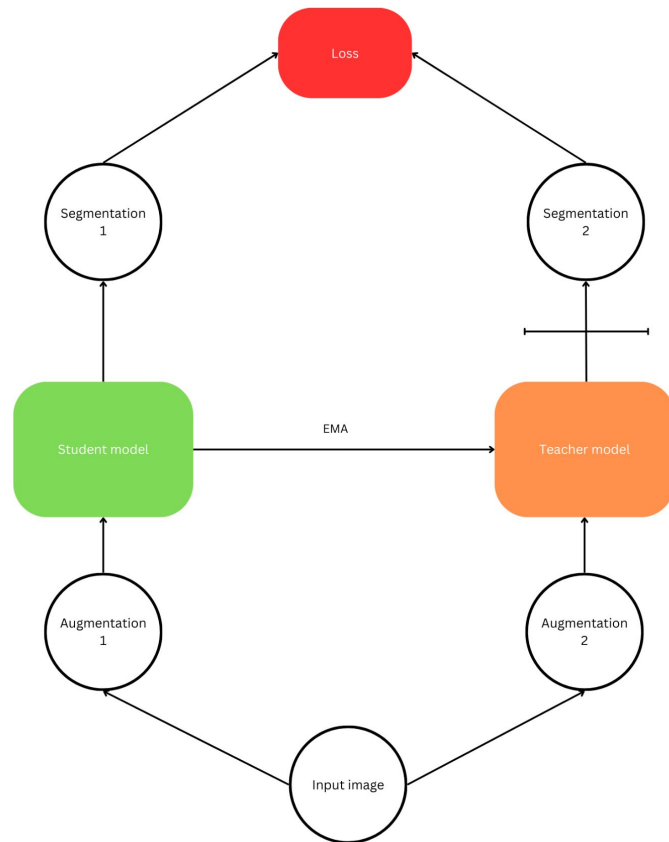
- **Teacher and student network:**

Teacher and student network compute a segmentation on two augmentations of the same image

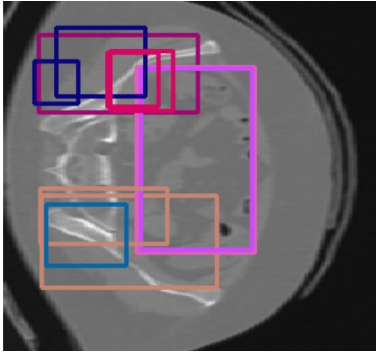
- **Training:**

Train the student by backpropagation of the loss.

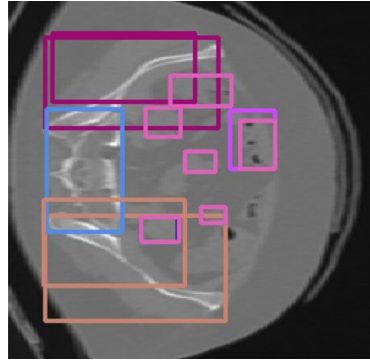
Update the teacher by Exponential Moving Average of the student.



# Results - Qualitative analysis

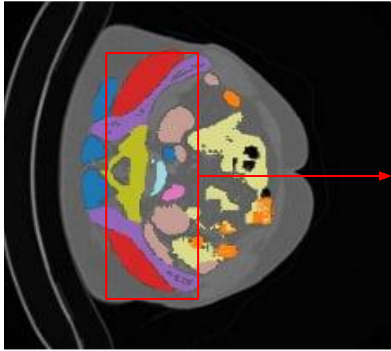


Yolo is trained to recognize a specific class into one box

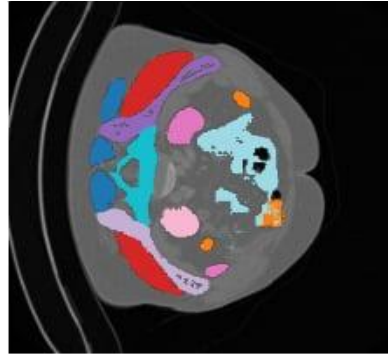


YOLO is trained to recognize the connected components of one class

# Results - Qualitative analysis



Yolo create big boxes in which SAM highlight all structures as the same class



YOLO is missing some boxes so SAM can't segment the associated structures

# Results - Quantitative analysis

| Method                   | Mean IoU | Dice Score |
|--------------------------|----------|------------|
| SAM2                     | 0.6      | 0.16       |
| SAM2 + YOLO              | 0.75     | 0.27       |
| Semi Supervised Learning | 0.76     | 0.30       |

- Score on private leaderboard : 0.3041 - Rank 7
- Improvement of segmentation with bounding boxes
- Dice score remains low, problem of detection and classification

# Results - Quantitative analysis

- Qualitatively satisfying segmentation outcomes
- Performance issues with complex object structures
- Limitations in post-processing and annotation exploitation may cap potential performance.

# Results - Strength and limitations

## **Strengths:**

- YOLO integration delivers high-quality bounding box prompts
- Effectively guides segmentation by targeting regions of interest

## **Limitations:**

- Incomplete utilization of available training annotations
- Insufficient post-processing leads to degradation on complex structures



# Results - Opportunities for improvements

## **Enhanced Post-Processing:**

- Implement advanced techniques to refine segmentation and bounding boxes outputs (not model based)

## **Better Annotation Utilization:**

- Explore dynamic loss weighting
- Improve bounding box extraction methods